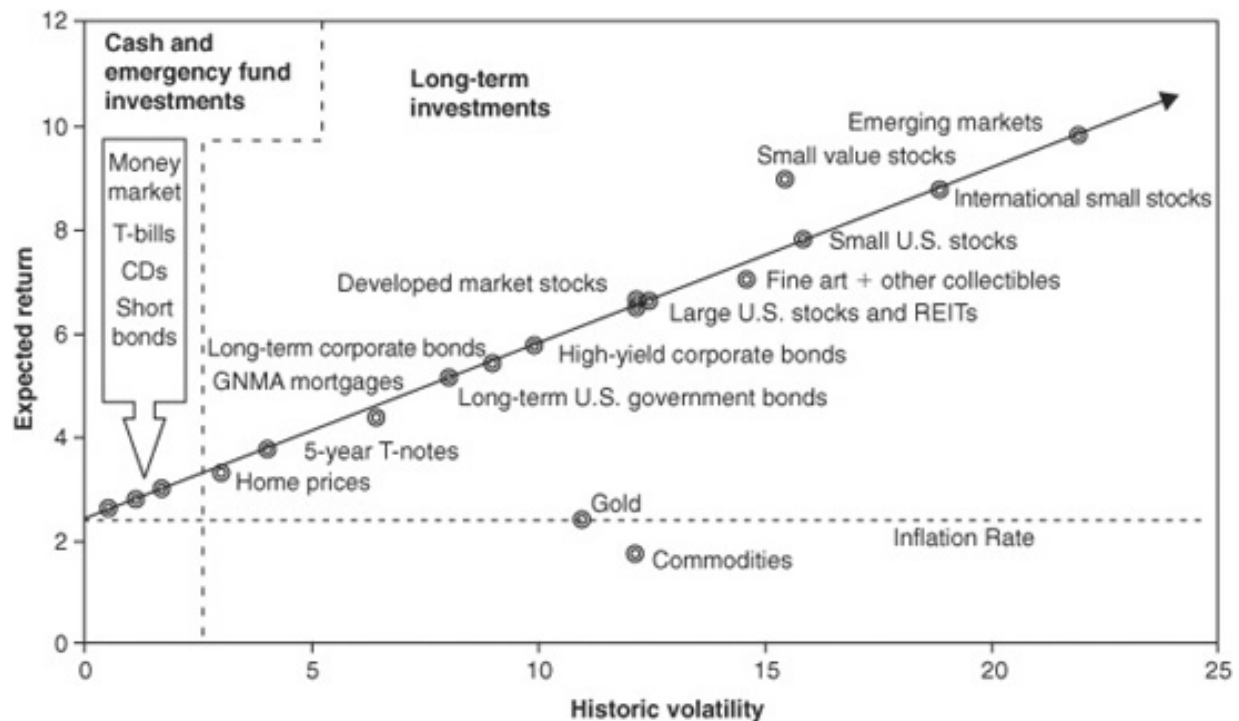




THE EFFECT OF INFLATION

The inflation rate is inherent in all market returns. Unfortunately, the returns generated by inflation are not “real” and do not buy extra goods and services. Nonetheless, our government taxes us as if inflation did add wealth.

Volatility-based models should factor out inflation expectations to derive a real return component for each asset class. [Figure 11-4](#) illustrates the effect that inflation has on annual Treasury bill returns. The top line is the nominal return on Treasury bills for a one-year period, and the darker line below is inflation-adjusted real return. The average one-year Treasury bill yield since 1955 has been 5.1 percent, less inflation this has averaged 3.8, leaving a real gain of 1.3 percent.

Taxes are also an issue. Over the long term, the inflation-adjusted rate of return on T-bills has averaged about 1.3 percent. However, income taxes are due on the entire 5.1 percent return. The government does not factor out inflation when calculating how much interest you earned even though it was government policies that created the inflation. The taxes on 5.1 percent interest are 1.3 percent, assuming a 25 percent tax rate. The real after-inflation, after-tax T-bills return was 0 percent—no return.